

Effective Programming Practices for Economists

Numerical Optimization

Derivative-Free Trust Region Algorithms

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Basic Idea (optimagic docs)

- Similar to derivative based trust region algorithm.
- Instead of Taylor approximation, use a surrogate model based on interpolation or regression.
 - Interpolation: Function is evaluated at exactly as many points as you need to fit the model.
 - Regression: Function is evaluated at more points than you strictly need. Better for noisy functions.
 - In general: Evaluation points are spread further out than for numerical derivatives.
- How the evaluation points are determined is complicated. It is also crucial for the efficiency of the algorithm.

Initial Evaluation



Actual = expected $\Rightarrow$ accept, increase trust region radius

Iteration 1



Actual = expected $\Rightarrow$ accept, increase trust region radius

Iteration 2



Actual $\approx$ expected, but step small $\Rightarrow$ accept, decrease trust region radius

Iteration 3



Actual $\approx$ expected, but step small $\Rightarrow$ accept, decrease trust region radius

Iteration 4



Actual $\approx$ expected, but step small $\Rightarrow$ accept, decrease trust region radius

Iteration 5



Converge because trust region radius shrinks to zero.

Some Remarks

- Within the trust region, the fit is generally better than the gradient based trust region algorithm
- By construction at the boundaries of the trust region for interpolation
- Choose between the two based on computation speed
 - If you have fast closed form derivatives, use the derivative based algorithm
 - If you only have numerical derivatives, use this instead
- It is intuitively very clear how this can work for noisy functions if enough evaluations are used for each surrogate model

A real algorithm: COBYLA

